

REFERRAL — RAVENSGATE, 19 NOVEMBER

SITE	RAVENSGATE DAM	OPERATOR	CAIRNHOLD POWER	DEATHS	2
POLICE FINDING	MURDER-SUICIDE	RDSI POSITION	DISSENT	AUTHOR	CHIEF INSPECTOR R. HALSEY

The Inspectorate does not investigate homicide and does not presume to. We investigate why plant kills people. On this occasion the two enquiries have produced findings that cannot both be correct, and the disagreement is not a matter of judgement. It is a matter of water.

The police investigation concluded that Ines Ballard killed Aurelio Sarn at 06:03 and then entered the forebay and drowned herself. Her body was recovered at 07:25 from **trash rack panel 4, inside the intake gallery**, at elevation 218.4 m.

THE POINT AT ISSUE

The intake gallery is not open water. It is a sealed concrete structure downstream of intake gate 2. It is dry above the sill whenever that gate is shut, and intake gate 2 was shut from 03:12:44 that morning until 08:40:19.

The police report does not address how the deceased came to be inside it. It records where she was found and does not ask how she got there. **We are unable to identify any route by which a body could enter the intake gallery after 03:12.**

We make no allegation and offer no alternative account. We say only that a finding which places a living woman at a switchgear cabinet at 06:02 must also explain how her body reached a location that was, by then, unreachable.

The Inspectorate's technical file is provided in full. Our own conduct in respect of the 2019 instrumentation renewal at this site is also within the scope of this review, and we have not redacted it.

SIGNED

R. HALSEY, CHIEF INSPECTOR

DATE

14 JAN

DEATHS OF A. SARN AND I. BALLARD

OPERATION	OP 24-11-2233	OPENED	19 NOV	CLOSED	08 DEC (19 DAYS)
SIO	DCI G. NAUNTON	FINDING	MURDER-SUICIDE	PERSONS ON SITE	6

SUMMARY OF FINDING

That Ines Ballard, Control Engineer, caused the death of Aurelio Sarn, Station Manager, by deliberately energising switchgear on which he was working, and thereafter took her own life by entering the forebay.

THE THREE LEGS

#	BASIS	SOURCE
1	Her credential. Cabinet C-14 was opened at 05:47:31 on credential I.BALLARD. Feeder tie F-3 was closed at 06:02:14 on credential I.BALLARD. Mr Sarn died at 06:03.	D-1
2	The argument. Ms Ballard and Mr Sarn were overheard at 02:25 in a serious disagreement in or adjacent to the control room. The subject was not established.	witness
3	No third party. The site was sealed by landslip from 19:40. Of the four other persons present, each was engaged in documented work at the material time, and none had reason to harm either deceased.	rosters, logs

MATTERS NOT RESOLVED

- Recorded here for completeness at the direction of the SIO.
- No note or communication of intent was recovered from Ms Ballard.
 - Ms Ballard's credential card was not on her person when recovered. Its location is recorded in the property schedule and was not thought material.
 - The Office of the Chief Medical Examiner was unable to estimate a time of death for Ms Ballard. The investigation proceeded on the basis that she died after Mr Sarn, that being the sequence the credential evidence implies.
 - No motive was established. Professional friction between the deceased was reported by two witnesses and denied by two others.

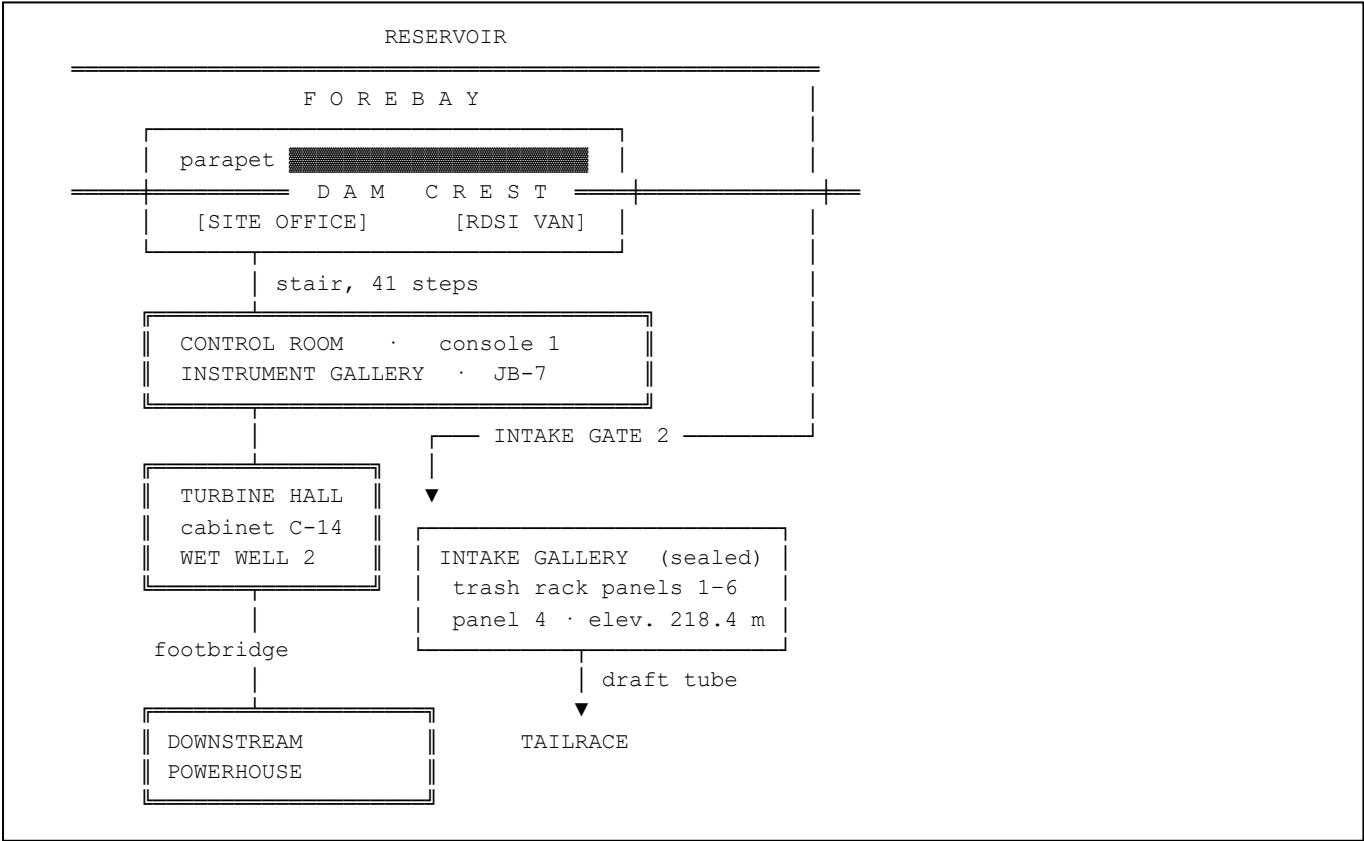
Note by the Case Review Unit. The third bullet above is the whole of the reason this file is in front of you. The sequence of the two deaths was not established by evidence. It was inferred from the finding, and the finding was then built on the sequence.

RAVENSGATE — ORIENTATION

PERSONS ON SITE, 18-19 NOVEMBER

NAME	POST	WHERE THEY WERE WORKING
SARN, Aurelio (58)	Station Manager	Came in for the storm. Mobile across the site. DECEASED 06:03
BALLARD, Ines (41)	Control Engineer	Control room and instrument gallery. DECEASED — time unknown
VEY, Halloran (47)	Shift Control Engineer	Console 1, control room
OKONKWO, Dessa (34)	Mechanical Maintenance Technician	Wet well 2, pump seal repair
WRENN, Piotr (61)	Dam Safety & Instrumentation Officer, RDSI	Instrument van, dam crest. On site for a scheduled inspection; stranded by the landslip.
MBEKI, Cato (26)	Assistant Operator	Downstream powerhouse

SITE PLAN — SCHEMATIC



WALKING TIMES — MEASURED BY THE REVIEW UNIT

FROM	TO	MINUTES
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Control room	Forebay parapet	4
Control room	Site office (across the crest)	5
Control room	Instrument gallery (41 steps)	3
Control room	Turbine hall, cabinet C-14	6
Downstream powerhouse	Anywhere on the main site	5 – footbridge only; service tunnel gated and padlocked for the storm

STORM CHRONOLOGY — UNCONTESTED

18 NOV 17:00 Storm Bettany. Night complement on site.

18 NOV 19:40 **Landslip severs the access road.** No person entered or left the site until 14:20 the following afternoon.

19 NOV Peak inflow to the reservoir.
04:00–05:00

19 NOV 06:10 Mr Sarn found in the turbine hall.

19 NOV 07:25 Ms Ballard found in the intake gallery.

POST-MORTEM — SARN, AURELIO

CASE	OCME 24-2871	AGE / SEX	58 / M	EXAMINER	DR H. ADEYEMI
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EXTERNAL EXAMINATION

Full-thickness contact burn to the right palm, 34 mm, with central charring and a raised margin — a classic electrical entry wound. Corresponding exit wound to the left forearm, 21 mm, at the point of contact with an earthed steel cabinet frame. Singeing of the forearm hair. No other injury of any kind. No defensive wounds. Nothing to suggest a struggle, and nothing to suggest restraint.

INTERNAL EXAMINATION

Findings consistent with a **hand-to-hand current path across the thorax** — the configuration that reliably arrests the heart. Contraction-band necrosis of the myocardium. No pre-existing cardiac disease of significance. No other contributory pathology.

TOXICOLOGY

Negative throughout. No alcohol, no drugs of abuse, no prescribed medication above therapeutic range, no carbon monoxide.

CAUSE OF DEATH

Ia. ELECTROCUTION — consistent with contact with an 11 kV conductor.

MANNER: UNLAWFUL KILLING. Death was effectively instantaneous.

TIME OF DEATH

06:03, 19 November. This is not a pathological estimate and does not need to be. The earth-fault alarm cascade at the station is timestamped 06:03, the deceased was found at 06:10, and the body was warm and unfixed on examination at the scene. **The time of this death is not in dispute and never has been.**

POST-MORTEM — BALLARD, INES

CASE	OCME 24-2872	AGE / SEX	41 / F	EXAMINER	DR H. ADEYEMI
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EXTERNAL EXAMINATION

Post-mortem abrasion and contusion to the anterior trunk, shoulders and knees, distributed in the pattern produced by prolonged contact with a coarse steel grating under flow. No injury shows a vital reaction — **all of it occurred after death.**

No defensive injuries. No grip marks to the arms. No ligature. No injury to the back of the head or neck. Fingernails intact and clean.

INTERNAL EXAMINATION

Frothy oedema fluid in the airway. Over-inflated lungs with rib impressions. Water in the stomach and duodenum. **Diatom recovery from bone marrow matches the forebay profile** and not the tailrace. She was alive when she entered the water and she entered it upstream.

CAUSE OF DEATH

Ia. DROWNING — freshwater, cold.

MANNER: OPEN. The pathology cannot distinguish between entering the water voluntarily, falling, and being put in. There is nothing on this body that decides it.

TIME OF DEATH — STATEMENT OF LIMITATION**READ THIS BEFORE RELYING ON ANY ESTIMATE**

The decedent was immersed in moving water at **3–5 °C** for an unknown but substantial period. This invalidates, individually and together:

Algor mortis	Body temperature equilibrated with the water long before recovery. The differential carries no information.
Rigor mortis	Cold profoundly retards onset and the body was additionally subject to continuous mechanical loading against the rack.
Livor mortis	Position was not fixed and flow was continuous. No usable pattern.
Vitreous potassium	Unreliable below 10 °C ambient. This Office does not report it in immersion cases.

THIS OFFICE CANNOT ESTIMATE A TIME OF DEATH FOR THIS DECEDENT AND DECLINES TO ATTEMPT ONE. Any sequence of the two deaths must be established from evidence other than pathology. It cannot be established from this report, and this report should not be cited as if it had been.

SCADA CORE EXTRACT

Master station log, 18 NOV 17:00 – 19 NOV 09:00. Time source: GPS-disciplined, synchronised throughout. Extract certified complete for the systems listed. Other subsystems (console session, alarm acknowledgement, permit and access) are held separately.

1 · SWITCHING & ISOLATION — TURBINE HALL

TIMESTAMP	EVENT	CREDENTIAL	INITIATED AT
19 NOV 05:47:31	CABINET C-14 — DOOR OPENED	I.BALLARD	LOCAL (cabinet)
19 NOV 05:52:06	FEEDER F-5 — ISOLATOR OPENED, LOTO PADLOCK APPLIED	A.SARN	LOCAL (switchroom)
19 NOV 05:53:44	FEEDER F-5 — PROVING TEST, DEAD CONFIRMED	A.SARN	LOCAL
19 NOV 06:02:14	FEEDER F-3 — TIE CLOSED	I.BALLARD	CONTROL ROOM, CONSOLE 1
19 NOV 06:03:02	EARTH FAULT — C-14 · CASCADE, 11 ALARMS	—	—
19 NOV 06:03:19	ALL 11 ALARMS ACKNOWLEDGED	H.VEY	CONTROL ROOM, CONSOLE 1

SYSTEMS NOTE SUPPLIED BY CAIRNHOLD POWER

Mr Sarn isolated feeder F-5, applied his personal lock, and proved the circuit dead. He did this correctly and the log records it. He was killed by feeder F-3.

2 · GATE OPERATIONS

TIMESTAMP	GATE	EVENT	INITIATED AT
19 NOV 02:31:08	INTAKE GATE 2	OPEN	CONTROL ROOM, CONSOLE 1
19 NOV 03:12:44	INTAKE GATE 2	CLOSED	CONTROL ROOM, CONSOLE 1
19 NOV 08:40:19	INTAKE GATE 2	OPEN (recovery operation)	CONTROL ROOM

Intake gates 1 and 3: not operated at any point during the storm.

Spillway gates: not operated. Overflow was passive throughout.

CAIRNHOLD POWER — SYSTEMS DISCLOSURE

SCADA CORE EXTRACT · RAVENSGATE STATION

D-1

SCADA/24-2233

PAGE 2 OF 2

3 · CREDENTIAL EVENTS — ALL SYSTEMS, 19 NOV 00:00-07:00

TIMESTAMP	CREDENTIAL	EVENT
01:50:12	P.WRENN	Control room — entry
01:58:40	P.WRENN	Control room — exit
02:12:05	D.OKONKWO	Confined space permit 1144 — entry, wet well 2
03:40:22	D.OKONKWO	Confined space permit 1144 — exit
05:35:41	D.OKONKWO	Confined space permit 1145 — entry, wet well 2
05:44:03	H.VEY	Control room — exit
05:46:12	H.VEY	Control room — entry
05:47:31	I.BALLARD	Cabinet C-14 — door opened
06:02:14	I.BALLARD	Feeder F-3 tie — closed, at console 1
06:05:51	C.MBEKI	Footbridge — crossing to main site
06:25:30	D.OKONKWO	Confined space permit 1145 — exit
No credential event of any kind attributed to I.BALLARD between 19 NOV 01:12 and 05:47:31.		

4 · RESERVOIR LEVEL & INFLOW

TIME	LEVEL M AOD	INFLOW M ³ /S	NOTE
19 NOV 00:00	241.10	88	Rising
19 NOV 02:00	241.44	119	Rising
19 NOV 02:31	241.52	126	Gate 2 opened
19 NOV 03:12	241.49	131	Gate 2 closed
19 NOV 04:12	241.71	184	Peak inflow
19 NOV 05:00	241.83	160	Falling
19 NOV 07:00	241.90	102	Falling

5 · SYSTEMS HELD SEPARATELY

Not included in this extract. Each is available to the review on request.

SYSTEM	HOLDER
Console 1 — alarm acknowledgement log	Cairnhold operations
Trash rack differential-pressure trace	Cairnhold civil
Property and credential audit (scene)	County Constabulary
As-built schedule amendment history	Cairnhold engineering records
Contract RG-0419 file, and payments thereunder	Cairnhold procurement

FILE F-1

CONTROL & OPERATIONS

Three documents. They are yours. Nobody else may read them until the Conference ends.

DOC	TITLE
C-1	Statement — H. Vey, Shift Control Engineer
C-5	Statement — C. Mbeki, Assistant Operator
D-1A	Console 1 — operator session record

Your brief. You speak for the control room. You know what an operator does during a storm, what the console records, and what it does not. When somebody at this table says "he was at his desk all night," you are the one who can say what that claim actually rests on.

H. VEY — SHIFT CONTROL ENGINEER

INTERVIEWEE	VEY, H. (47)	DATE	19 NOV, 16:20	STATUS	WITNESS
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DS Take me through the night.

HV I took console at ten. Bettany was already on us. During a storm you don't leave the desk — the alarms come in threes and fours and you're acknowledging, you're watching levels, you're deciding whether to pass water. I was at console one from twenty-two hundred until Cato came running in at ten past six.

DS Continuously?

HV Continuously. That's the job.

DS You opened intake gate 2 at half past two.

HV I did. Inflow was up to a hundred and twenty-six and climbing and I wanted to take the top off before the peak came through. Forty minutes of gate two, then I shut it. You'll find that's textbook and you'll find the Inspectorate said so.

DS Tell me about Ines Ballard.

HV Ines was the best engineer on this station and I'd say that to anyone. She was also somebody who got hold of a thing and would not put it down. She'd got hold of the foundation piezometers — said the numbers weren't moving with the reservoir. I said, it's a storm, look at it Thursday.

DS Did she look at it Thursday?

[Pause.]

HV No. She went down the gallery with Aurelio at about ten past two.

DS And then?

HV And then there was an argument. Twenty-five past two, in the corridor. Raised voices, the pair of them. I stayed at my desk. It isn't my place and I had a storm on.

DS Did you see either of them after that?

HV Aurelio, at twenty to six. He came in and I told him Ines had gone off somewhere and that C-14 was throwing an earth fault on F-5. He said he'd look at it. That's the last thing I said to him.

DS Where was Ms Ballard?

HV I assumed the site office. She wanted the as-builts. I thought she'd gone to dig them out and stew about it, frankly.

DS Her card opened C-14 at quarter to six.

HV Then she was there, wasn't she.

[Interviewee becomes distressed. Break taken 16:51-17:04.]

HV Forty-one years old. I have known her since she was a graduate. Whatever she did, I would like somebody to remember that.

C. MBEKI — ASSISTANT OPERATOR

INTERVIEWEE	MBEKI, C. (26)	DATE	19 NOV, 13:05	STATUS	WITNESS — FOUND
					DECEASED 1

DS Where were you?

CM Downstream powerhouse. Nine hours. Aurelio put me over there at nine to babysit the number two set — it was hunting on load and somebody had to sit with it through the storm. You can't leave a machine like that on its own.

DS Nine hours alone.

CM Across a footbridge, in a gale, at night, on my own, yes. It was the worst night of my life and then it got worse.

DS Did you cross back at any point before six?

CM No. You badge the footbridge door both ways, it's all on the system. I crossed at four minutes past nine and I crossed back at five past six.

DS Tell me about six.

CM The set tripped. Whole board went. I crossed back and went up to the turbine hall because that's where the fault was showing, and he was —

[Pause.]

CM He was on the floor by C-14. His lock was still on the F-5 isolator. I want to say that because people are saying he was careless and he wasn't. His padlock was on. He'd tested it. He did it right.

DS You held a spare control room key.

CM In September, for two weeks, when the lock was changed. I gave it back. It's in the register.

DS Anything else?

CM Only that Ines was fine. People keep asking me if she seemed like somebody who — and she was fine. She was cross about the piezometers. Being cross about instruments is not the same as anything else and I wish somebody would write that down.

[Recorded as stated.]

CONSOLE 1 — SESSION RECORD, 18-19 NOVEMBER

Operator logged on	H.VEY, 18 NOV 21:58:11
Operator logged off	19 NOV 07:41:02
Session interruptions	NONE
Second operator logged on at console 1	NONE
Console lock events	NONE
Console 2 (backup) session	NOT LOGGED ON AT ANY TIME

A single unbroken operator session of nine hours and forty-three minutes. No other operator logged on to console 1 or console 2 at any point that night. Every control action taken at console 1 during the storm was taken inside this session.

CONTROL ACTIONS TAKEN AT CONSOLE 1

TIME	ACTION
22:14	Unit 1 load setpoint adjusted
23:40	Unit 2 governor droop adjusted
01:05	Unit 1 load setpoint adjusted
02:31	Intake gate 2 — OPEN
03:12	Intake gate 2 — CLOSED
04:20	Unit 2 load setpoint adjusted
06:02	Feeder F-3 tie — CLOSED (credential I.BALLARD)

NOTE ON CREDENTIALS AT A CONSOLE

A logged-on operator session and a credentialled control action are separate records. The session says who is *signed in*. A control action on a permissioned circuit additionally requires a card to be presented at the console reader at the moment of the action. **The two need not be the same person, and the system does not require them to be.**

Operations note: this record shows only that the session was open. It does not, and cannot, show whether the operator was in the chair.

FILE F-2

HYDROLOGY & RECOVERY

Three documents. They are yours. Nobody else may read them until the Conference ends.

DOC	TITLE
D-2	Recovery report — I. Ballard
D-2A	Intake gallery — hydrological statement & section
D-2B	Forebay and gallery — flow conditions

Your brief. You speak for the water. The Inspectorate's referral note on the Case Board asks one question and it is your question: **how did she get in there?** Nobody else at this table can answer it. Decide when to.

RECOVERY — I. BALLARD, INTAKE GALLERY

LOCATED

19 NOV 07:25

RECOVERED

19 NOV 09:14

OFFICER

SGT A. LINDQVIST

LOCATION

Structure	INTAKE GALLERY (SEALED)
Position	TRASH RACK PANEL 4
Elevation of body	218.4 m AOD
Relative to intake gate 2	DOWNSTREAM
Gallery condition on entry	DRY ABOVE SILL (211.0 m)

NARRATIVE

Search of the forebay by boat and pole was negative from first light. At 07:25 the gallery was entered through the upstream inspection hatch on the instruction of the duty civil engineer, who observed that if the deceased were not in open water she could only be against the racks.

The deceased was found on the upstream face of trash rack panel 4, in a semi-prone position, held against the bars by nothing but her own weight and the shape of the panel. The gallery was dry above the sill and she was not in water at the time of recovery.

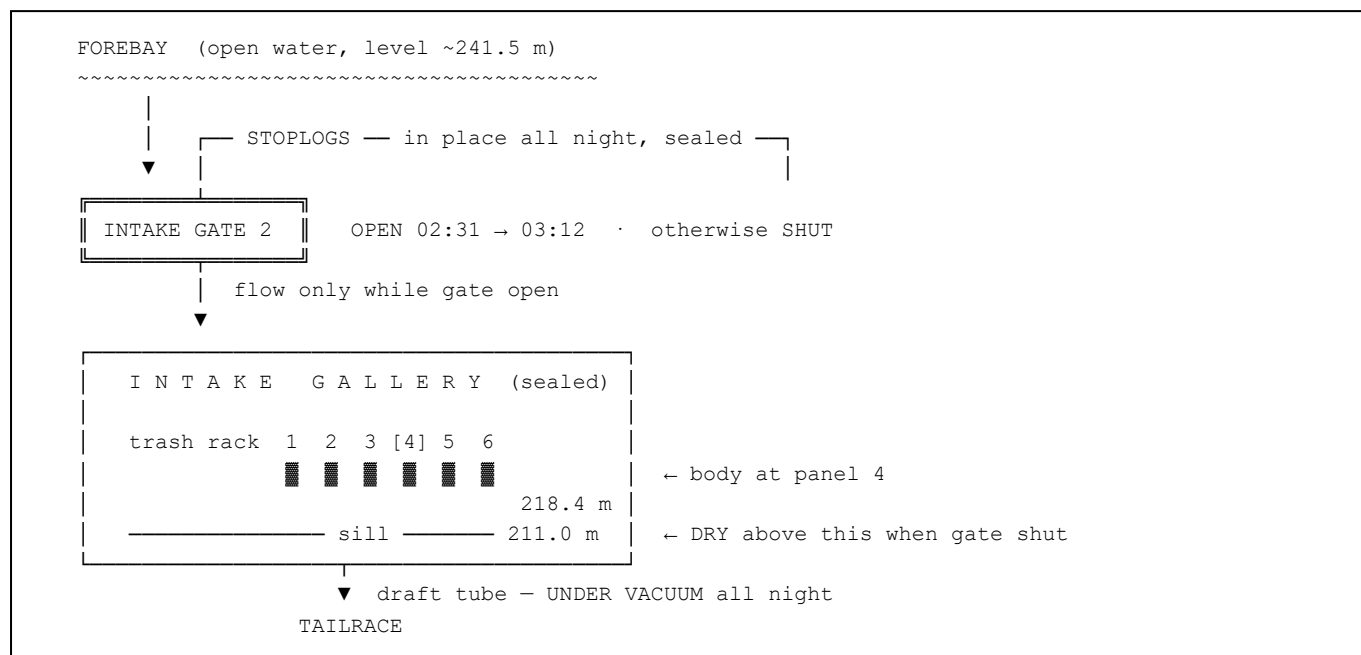
The deceased lay beneath a consolidated mat of storm debris — bracken, spruce needles, and a length of fence rail — packed hard against the rack and extending across panels 3 to 5. The mat was cut away to reach her. Approximately 40 kg of material was removed.

The Unit does not offer an opinion on when this material arrived. Cairnhold civil hold the rack differential-pressure trace, from which the question can be answered.

OBSERVATIONS

- Clothing intact, boots laced, hi-vis fastened. No sign of preparation to enter water.
- A permanent marker was still clipped in the chest pocket.
- **No credential card and no lanyard on the body or in any pocket.**
- Wristwatch stopped. Face broken by rack contact. The Unit places no weight on the stopped time and has not recorded it, immersion having continued long after the impact.

HOW WATER — AND ANYTHING IN IT — REACHES PANEL 4



STATEMENT

1. **The gallery is sealed.** There are three openings: intake gate 2, the upstream inspection hatch, and the draft tube. There are no others.
2. **The inspection hatch was dogged shut** from the storm preparation on 18 November at 16:40 until it was opened by the recovery team at 07:25 on 19 November. It is bolted from outside. It cannot be opened from within the gallery.
3. **The draft tube ran under vacuum** continuously while unit 1 was generating, which it was all night without interruption. Flow through the draft tube is *outward and downward* at all times. Nothing enters the gallery this way. Nothing can.
4. **The upstream stoplogs were in place** and sealed for the duration of the storm, as logged in the storm-preparation checklist.
5. **Therefore the only route from the forebay to trash rack panel 4 is through intake gate 2, while intake gate 2 is open and flowing.**
6. **Whenever intake gate 2 is shut, the gallery is dry above the sill at 211.0 m.** The body was at 218.4 m — seven metres above the sill. There is no standing water at that elevation with the gate shut. There is nothing there at all.

Conclusion of this statement. The deceased passed through intake gate 2 while it was open and flowing. Cairnhold Civil offers no view on how she came to be in the forebay. We say only that she was carried into the gallery by water, and that the water was only there for as long as the gate was.

SIGNED

T. ARGYLE, PRINCIPAL CIVIL ENGR

DATE

02 DEC

TRANSIT AND RETENTION

TRANSIT

Forebay surface to intake gate 2 mouth	~40 m
Surface velocity toward the intake, gate 2 open	0.6-0.9 m/s
Approximate transit, parapet to rack	2 to 4 minutes
Transit with gate 2 shut	NO FLOW — NO TRANSIT

A body entering the forebay near the parapet while gate 2 is open will be drawn to the racks within a few minutes. The intake draws from the surface layer during a spill of this magnitude, and floating objects are recovered from the racks routinely after every storm.

RETENTION

Once flow ceases, an object on the racks stays on the racks. There is no mechanism to move it. The gallery is dry, still and enclosed. This is why the racks must be raked by hand after every significant event.

RAKE RECORD — TRASH RACK, INTAKE GALLERY

DATE	PANELS RAKED	RECOVERED	BY
02 NOV	1-6	Vegetation, 60 kg. Nothing else.	D. Okonkwo
17 NOV	1-6	Vegetation, 20 kg. Racks left clear.	D. Okonkwo
19 NOV	—	Recovery operation. See D-2.	

The racks were raked clear on 17 November, the day before the storm. Everything found on them on 19 November arrived during Storm Bettany.

Cairnhold Civil holds a continuous **differential-pressure trace** across the racks. Debris loading is directly readable from it, including the moment any significant mat formed. It is not reproduced here.

FILE F-3

MECHANICAL & MAINTENANCE

Three documents. They are yours. Nobody else may read them until the Conference ends.

DOC	TITLE
C-3	Statement — D. Okonkwo, Mechanical Maintenance Technician
C-3A	Confined space permits & atmospheric monitor telemetry
E-6	Cabinet C-14 — plant report & isolation procedure

Your brief. You speak for the plant and the people who work on it. You are the only person at this table who can explain what a lockout actually is, and therefore the only one who can say why a man who did everything right is dead.

D. OKONKWO — MECHANICAL MAINTENANCE TECHNICIAN

INTERVIEWEE	OKONKWO, D. (34)	DATE	20 NOV, 10:30	STATUS	WITNESS
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DS Where were you between two and four?

DO Down wet well 2 replacing a mechanical seal on the drainage pump. Permit 1144. You don't go into a confined space without a permit and a monitor, and the monitor talks to the control room the whole time you're down there. It's all logged, minute by minute. I was in at twelve past two and out at twenty to four.

DS And later?

DO Back down at twenty-five to six. Permit 1145. Out at twenty-five past. I came up into the middle of it.

DS You filed a grievance against Mr Sarn in August.

DO I did. He put me on nights for a month without consultation and I said so through the proper channel and I lost. That is the whole of it. I liked Aurelio. Losing a grievance to somebody is not a reason to — you're not seriously —

[Interviewee declines to complete the sentence.]

DS You signed the goods-receipt for the piezometer array in 2019.

[Long pause.]

DO I did.

DS Did you check it?

DO No.

DS Why not?

DO Because it was hour ten of a twelve, because Halloran put the sheet in front of me and said the Thennick lads have been and gone and it's all in, and because he is a chartered engineer and I am a fitter and that is how it works here. I signed for six instruments I never saw. Nobody has to tell me what that means. I have had six weeks to think about what that means.

DS Did anyone ever raise it with you?

DO Not once in five years. Not the regulator, not the auditors, nobody. And I'll tell you what I've realised since — if those instruments had ever been read by anyone who cared what the numbers meant, somebody would have noticed. So either nobody ever looked, or somebody was looking at numbers that came from somewhere else.

WET WELL 2 — ENTRIES OF 19 NOVEMBER

PERMIT 1144

Entrant	D. OKONKWO
Authorised by	A. SARN (Station Manager)
Entry	02:12:05
Exit	03:40:22
Task	DRAINAGE PUMP 2 — MECHANICAL SEAL
Standby attendant	NONE — REMOTE MONITORING

PERMIT 1145

Entrant	D. OKONKWO
Entry	05:35:41
Exit	06:25:30
Task	SEAL BEDDING-IN CHECK, RUN TEST

HOW THE MONITOR WORKS — AND WHY THIS IS NOT A SIGNATURE ON A SHEET

A personal atmospheric monitor is issued against the permit and worn on the entrant. It transmits **every fifteen seconds** to the control room for as long as it is below the wellhead — oxygen, hydrogen sulphide, methane, and a received-signal figure that falls away sharply with depth. The telemetry is what tells the surface that somebody is still down there and still breathing.

It cannot be left at the bottom of the well and walked away from. The signal profile of a stationary monitor and a worn one are not alike, and the unit alarms on thirty seconds of no motion.

TELEMETRY SUMMARY

PERMIT	TRANSMISSIONS EXPECTED	RECEIVED	GAPS	MOTION ALARMS
1144	354	354	NONE	NONE
1145	199	199	NONE	NONE

Unbroken worn-monitor telemetry from 02:12:05 to 03:40:22, and again from 05:35:41 to 06:25:30. Wet well 2 is beneath the turbine hall floor, reached by a single fixed ladder through one hatch.

CABINET C-14 AND THE ISOLATION THAT FAILED

CONDITION ON EXAMINATION

Cabinet door	OPEN, undamaged
F-5 isolator	OPEN — PADLOCK APPLIED, TAG SIGNED A.SARN
Proving unit	ON THE FLOOR, FUNCTIONAL, TESTED CORRECT
F-3 tie	CLOSED
Arc damage	MINOR, AT THE POINT OF CONTACT

LOCKOUT-TAGOUT, IN PLAIN TERMS

Before working on switchgear, you isolate the circuit at the breaker, apply **your own** padlock to the isolator, hang your tag on it, and then prove the circuit dead with a tester you have first proved on a live source. Nobody can remove your lock. Nobody can re-energise past it. It is a good procedure and it has kept people alive for eighty years.

It protects the circuit you locked. It says nothing whatever about a second circuit you did not know existed.

FINDING OF THIS REPORT

Mr Sarn did everything correctly.

He isolated feeder F-5. He applied his personal padlock and his signed tag. He proved the circuit dead with a tester that this department has since examined and found accurate. The cabinet was dead when he opened it and he had proved it so.

He was then killed at 06:03 by **feeder F-3**, a second supply to the same busbar, closed remotely from the control room at 06:02:14.

Question this department cannot answer. The as-built schedule for C-14 held in the control room — the drawing an engineer consults before isolating — shows this cabinet as **single-fed from F-5**. It does not show F-3. Whether that is an error in the drawing, or something else, is outside our competence and outside this report. Cairnhold engineering records hold the amendment history.

FILE F-4

SITE SECURITY & PERSONNEL

Three documents. They are yours. Nobody else may read them until the Conference ends.

DOC	TITLE
E-1	Door logs — control room and footbridge
E-1A	Roster, shift pattern & key issue register
C-5A	Credential practice at Ravensgate — witness note

Your brief. You speak for who was where, and for what a badge actually proves. The entire original finding rests on a credential. You hold the papers that say what a credential is worth at this site.

CONTROL ROOM & FOOTBRIDGE — 18-19 NOVEMBER

CONTROL ROOM DOOR — ALL EVENTS

TIMESTAMP	DIR	CREDENTIAL
18 NOV 21:57:02	IN	H.VEY
18 NOV 23:11:40	IN	I.BALLARD
19 NOV 01:12:18	OUT	I.BALLARD
19 NOV 01:50:12	IN	P.WRENN
19 NOV 01:58:40	OUT	P.WRENN
19 NOV 02:24:31	IN	A.SARN
19 NOV 02:29:55	OUT	A.SARN
— no control room door event of any kind, 02:29:55 → 05:44:03 —		
19 NOV 05:44:03	OUT	H.VEY
19 NOV 05:46:12	IN	H.VEY
19 NOV 06:10:44	IN	C.MBEKI

Between 02:29:55 and 05:44:03 — three hours and fourteen minutes — the control room door did not open. H. Vey was inside it, having entered at 21:57 and not left. No other person entered. Whatever was in that room during those three hours was accessible to one person and to nobody else.

Security note: the control room door is badge-controlled on entry and on egress. There is one door. The fire escape is alarmed and did not activate.

FOOTBRIDGE DOOR — DOWNSTREAM POWERHOUSE CROSSING

TIMESTAMP	DIR	CREDENTIAL
18 NOV 21:04:33	OUT	C.MBEKI (to powerhouse)
— no activation, 21:04:33 → 06:05:51 —		
19 NOV 06:05:51	IN	C.MBEKI (to main site)

The footbridge is the only route between the downstream powerhouse and the main site. The service tunnel was gated and padlocked as part of storm preparation on 18 November at 16:20 and was found still padlocked on 19 November. **C. Mbeki was on the far side of the river from 21:04 until 06:05.**

CAIRNHOLD POWER — SITE SECURITY

ROSTER, SHIFT PATTERN & KEY REGISTER

E-1A

F-4

PAGE 1 OF 1

PERSONNEL — STORM WATCH, 18-19 NOVEMBER

NAME	ON DUTY	ASSIGNED STATION	CREDENTIAL NO.
SARN, A.	17:00 → (open)	Mobile — storm command	0001
VEY, H.	22:00 → 08:00	Control room, console 1	0004
BALLARD, I.	22:00 → 08:00	Control room, secondary	0007
OKONKWO, D.	18:00 → 06:00	Mechanical, mobile	0022
MBEKI, C.	21:00 → 07:00	Downstream powerhouse	0031
WRENN, P.	visitor	RDSI instrument van, crest	V-118

KEY ISSUE REGISTER — CONTROL ROOM

ISSUED TO	REASON	RETURNED
04 SEP C. MBEKI	Lock change — temporary override key	18 SEP — SIGNED IN BY A. SARN
—	No other physical key to the control room has been issued in three years. Access is by credential.	

CREDENTIAL POLICY — EXTRACT, SITE STANDING ORDERS §11

11.2 A credential card is personal and shall be carried on the person at all times while on site.
11.3 A credential card shall not be lent, left unattended, or used by any person other than the holder.
11.4 The system records the card. Responsibility for a recorded event rests with the holder.

Standing Orders are what the site says it does. Whether it does it is a separate question, and one this file has a note on.

WHAT A BADGE PROVES HERE

Compiled by the Review Unit from four statements. The original investigation did not take this point and did not need to, having reached its finding by another route.

- W-1 "Nobody wears the card on the crest in a gale. It's on a lanyard, it blows about, it catches. You leave it on the desk and you pick it up when you come back in. Everybody does it. Everybody has always done it."
- W-2 "Ines was the worst for it. She'd swipe in, drop the lanyard on the desk, and be off down the gallery. I've handed it back to her I don't know how many times."
- W-3 "There was a memo about it. There's a memo about everything."
- W-4 "You can't badge into the control room and then take your card out to the crest, because then you can't get back in. So you leave it. That's not laziness, that's the door."

DOCUMENTED	
Site memo SO-11 re-issued	MAR and AUG, same year
Informal warnings recorded, I. BALLARD, §11.2/11.3	TWO
Formal disciplinary action, any person, §11	NONE, EVER
Card readers requiring a second factor	NONE
Card readers with a camera	NONE

Note by the Review Unit. Standing Order 11.4 says responsibility for a recorded event rests with the holder. That is a rule about discipline. It is not a finding of fact about who was standing at the reader, and it cannot be used as one.

Where Ms Ballard's card actually was that night is a question of physical evidence. The scene property schedule will answer it. This file does not hold the property schedule.

FILE F-5

REGULATOR — RDSI

Three documents. They are yours. Nobody else may read them until the Conference ends.

DOC	TITLE
E-2	RDSI inspection — instrument gallery, junction box JB-7
E-2A	Contract RG-0419 — scope and certification history
C-4	Statement — P. Wrenn, Dam Safety & Instrumentation Officer

Your brief. You speak for the regulator — including for its failures. You hold the reason all of this happened, and one of the people holding it is your own officer. Your file is the only one that explains *why*. It says almost nothing about *who*.

JUNCTION BOX JB-7 — FOUNDATION PIEZOMETER ARRAY

INSPECTED

21 NOV

INSPECTOR

C. FALCONE, RDSI

OFFICER EXCLUDED

P. WRENN
(CONFLICT)

FINDING

Piezometers P-7 to P-12 do not exist.

Junction box JB-7, interior	EMPTY
Cabling installed under RG-0419	TERMINATES UNCONNECTED BEHIND THE BOX
Borehole collars P-7 to P-12	NOT PRESENT. NO DRILLING EVIDENCE.
Instruments recorded as installed 2019	6
Instruments actually present	0
Values reported against P-7 to P-12 since 2019	CONTINUOUS, PLAUSIBLE, COMPLETE

The array was never installed and has been reporting for five years. The values submitted to this Inspectorate quarterly since 2019 were generated by something other than an instrument in the ground.

Ms Ballard's observation on the night — that the reported values did not respond to reservoir level — is correct, and is the signature of synthetic data. A real foundation piezometer under a rising reservoir cannot fail to move.

CONSEQUENCE

The actual foundation pore pressures at Ravensgate are unknown and have been unknown for five years. The Inspectorate has issued a Section 10 direction. The station is operating under restriction pending a full drilling and instrumentation programme.

This is a matter of very considerable regulatory gravity independent of the two deaths, and the Inspectorate wishes it recorded that it is not a small thing that happened alongside a murder. It is the reason there was one.

CONTRACT RG-0419

TITLE	FOUNDATION INSTRUMENTATION RENEWAL & LEGACY FEEDER DECOMMISSIONING	CONTRACTOR	THENNICK INSTRUMENTATION LTD
LET	2019	VALUE	\$1.41 M

SCOPE — TWO PARTS

#	SCOPE	CERTIFIED COMPLETE	ACTUALLY DONE
A	Install foundation piezometer array P-7 to P-12, instrument gallery, with junction box JB-7 and telemetry	YES — 2019	NO (E-2)
B	Permanently decommission and physically remove legacy 11 kV feeder F-3, turbine hall	YES — 2019	NOT KNOWN TO THIS INSPECTORATE

NOTE BY THE INSPECTORATE, ADDED 12 JAN

Scope A was certified complete and was not done. Scope B was certified complete by the same signature on the same day under the same contract. **The Inspectorate has no basis to assume scope B was executed and every reason to doubt it.**

We note without comment that Mr Sarn is recorded as having been killed by feeder F-3.

CERTIFICATION & RECEIPT

Cairnhold certifying engineer, scopes A and B	HELD IN CAIRNHOLD ENGINEERING RECORDS — NOT SUPPLIED TO RDSI
Goods-receipt signatory, scope A	D. OKONKWO
RDSI acceptance of the 2019 data package	P. WRENN — DOCUMENT REVIEW ONLY, NO SITE VERIFICATION

RDSI procedure permits acceptance on document review where the certifying engineer is chartered. It is permitted. On this occasion it was catastrophic.

P. WRENN — DAM SAFETY & INSTRUMENTATION OFFICER, RDSI

INTERVIEWEE	WRENN, P. (61)	DATE	21 NOV, 09:00	STATUS	WITNESS
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DS Why were you on site?

PW Scheduled instrument inspection, booked in September. I'd have been away by seven if the hill hadn't come down on the road. I spent the night in the van with a laptop and a flask like a man half my age.

DS You went into the control room at ten to two.

PW For eight minutes, to ask Halloran whether he intended to pass water, because if he did I wanted to be logging while he did it. He said he would at half past. I went back to the van and I logged it.

DS Did you see Ms Ballard?

PW Not that night. She telephoned the van at about one, wanting to know whether I had the 2019 commissioning data for the foundation array, and I said it would be in the office in the morning.

DS Did she say why she wanted it?

PW She said the numbers were dead. Those were her words. "Piotr, the numbers are dead, they haven't moved in six hours and the reservoir's come up a third of a metre."

DS What did you say?

[Pause.]

PW I said instruments drift and we'd look at it Thursday.

DS You accepted the 2019 data package for that array.

PW I did.

DS Without a site verification.

PW Without a site verification. It is permitted where the certifying engineer is chartered and Halloran Vey is chartered. And I will tell you the true reason, because you will find it anyway and I would rather say it. The gallery is down forty-one steps and I was three weeks off a hip replacement and I did not want to go down them. That is the whole of my defence and it is not one.

DS You and Mr Vey worked together.

PW At Thennick. Fifteen years ago, for about two. He was good. I'd have said he was one of the better ones.

DS Where were you at six this morning?

PW On the satellite phone to my duty desk telling them we had a fatality, and they record those calls, and I would be obliged if you would go and listen to it, because a man is dead on a station I signed off and I am sixty-one years old and I would like somebody to check.

CASE REVIEW — SARN AND BALLARD (DECEASED)

On the night of 18 November, Storm Bettany closed the road to Ravensgate Dam and six people were sealed in with each other until the following afternoon. By dawn two of them were dead.

Aurelio Sarn, the Station Manager, was electrocuted in the turbine hall at 06:03. Ines Ballard, a Control Engineer, was found drowned at 07:25 in the intake gallery. Ms Ballard's credential card had opened the cabinet Mr Sarn died at, and had closed the circuit that killed him.

The police concluded murder–suicide and closed the file in nineteen days.

The Reservoirs & Dam Safety Inspectorate does not agree, on a technical point about water, and has said so in writing. This review is convened because two competent authorities have reached findings that cannot both be true.

YOUR TASK

You are the review team. Between you, you hold every file both enquiries generated — but no one of you holds all of it.

Establish what happened at Ravensgate. If the original finding is wrong, say so, say who, and show your working.

WHAT IS DIFFERENT ABOUT THIS CASE

The file is split. The Case Board in the middle of the table is public and belongs to everybody. Everything else is in five Specialist Files, and each of you holds one. You may not read another detective's file. You may not pass it across. You may only **table** what you choose to table, when you choose to.

This is not a game of hidden traitors. Nobody at this table is lying to anybody. Every one of you wants the same answer. The difficulty is that the answer is in five pieces and the pieces do not know about each other.

A WARNING ABOUT A DEAD WOMAN

This review begins with a conclusion already reached about Ines Ballard, who is not here to argue with it. Nineteen days of police work say she killed a man and then drowned herself. If you decide they were right, say so — that is a legitimate finding and you may write it down.

Just be sure it is a finding and not an inheritance.

THE CASE CONFERENCE

1 · WHO HOLDS WHAT

DETECTIVES ALLOCATION OF THE FIVE SPECIALIST FILES	
2	F-1 + F-3 / F-2 + F-4 + F-5 (two Sessions, not three)
3	F-1 + F-3 / F-2 / F-4 + F-5
4	F-1 / F-2 / F-3 + F-4 / F-5
5	One file each
6	One file each, and a sixth detective as CASE OFFICER — holds no file, keeps the Board and the Resolution Sheet, and may query but never table

Solo play does not use the Conference. See the reverse of the Resolution Sheet.

2 · THREE SESSIONS

The review proceeds in **three Sessions**. Take a breath between them; they are not timed.

In each Session, each detective may TABLE one document from a file they hold. To table it: read it aloud, then lay it face up on the Case Board. It is public from that moment and stays public.

One document. Per detective. Per Session. Choose it well — and notice that what is worth tabling in Session 3 is not what was worth tabling in Session 1.

3 · THE QUERY

At any moment, any detective may **QUERY**: name a subject, a piece of plant, a person or a clock time.

Every other detective must answer truthfully whether their file holds anything on it. Yes or no. They need not say what it is, and tabling it still costs a Session slot.

Queries are free and unlimited. Asking the right question is the skill this case is testing.

4 · AUTHORITIES

Between Sessions — never during one — the table may spend from a shared pool of Authorities, exactly as in Case 01. Twelve lines of inquiry are listed overleaf.

DETECTIVES	AUTHORITIES
1 – 2	5
3 – 4	6
5 – 6	7

5 · AFTER SESSION 3

All files open. Everything is shared, everyone reads everything, and the review reaches its finding with the whole file in front of it.

But: each step of the Case Resolution Sheet **scores double** if it was established on the Case Board *before the end of Session 3*. Reaching the right answer after everything is face up is worth something. Reaching it while the file was still in pieces is worth twice as much.

6 · ACCUSING

Two wrong accusations are permitted; each costs 10 points and returns a **Reconsider** envelope. **Your third accusation is final.** Then, and only then, open **Envelope S-1**.

OPEN LINES OF INQUIRY

Twelve authorities. Each may be exercised once. Spend only between Sessions. Tick the box, then open the matching sealed slip.

SPENT NO.	AUTHORITY	DIRECTED TO
☐ 01	Obtain the console 1 alarm acknowledgement log	Cairnhold operations
☐ 02	Scene property and credential audit — where was each card recovered?	County Constabulary
☐ 03	Re-interview C. Mbeki regarding the spare control room key	CID
☐ 04	Trash rack differential-pressure trace and debris analysis	Cairnhold civil
☐ 05	Second post-mortem on I. Ballard, independent pathologist	OCME
☐ 06	Financial disclosure — H. Vey and associated entities	Financial Investigation Unit
☐ 07	As-built amendment history and the RG-0419 decommissioning certificate	Cairnhold engineering records
☐ 08	Recover the 02:25 argument — who overheard it, and what was said	CID
☐ 09	Full movement reconciliation for P. Wrenn (RDSI officer)	CRU
☐ 10	Metallurgical examination of the C-14 contact surfaces	Forensic science
☐ 11	Thennick Instrumentation Ltd — corporate and delivery records	Companies registry
☐ 12	Forebay parapet examination — fibre, contact traces, damage	Forensic science

AUTHORITIES REMAINING

☐

☐

☐

☐

☐

☐

☐

Use only as many boxes as your table is entitled to — five, six or seven. There are twelve authorities and you cannot have half of them.

ADVICE FROM THE UNIT

Four of these twelve return nothing you do not already have. They are not traps; they are the ordinary shape of an investigation, and the skill is telling them apart from the others *before* you pay. Every authority worth buying was suggested by a document somebody has already tabled. If nobody at the table can point to that document, do not buy it yet — **query first**. Queries are free.

CASE RESOLUTION SHEET

Cite by document reference (B-2, D-2A, SLIP 04). Tick **S≤3** if the step was established on the Case Board before the end of Session 3 — those steps score double.

1. 1 · THE TWO CAUSES OF DEATH

Evidence S≤3

2. 2 · WHEN DID INES BALLARD ENTER THE WATER — AND BY WHAT ARGUMENT?

Evidence S≤3

3. 3 · WHAT DOES A CREDENTIAL EVENT ACTUALLY PROVE?

Evidence S≤3

4. 4 · MECHANISM — WHY DID A CORRECT ISOLATION FAIL TO PROTECT HIM?

Evidence S≤3

5. 5 · OPPORTUNITY — WHO COULD HAVE USED HER CARD, AND FROM WHERE?

Evidence S≤3

6. 6 · CONTRADICTION — WHOSE ACCOUNT OF THAT NIGHT FAILS?

Evidence S≤3

PERSONS

PERSON	FINDING	EVIDENCE RELIED ON
BALLARD, Ines named by the original finding		
VEY, Halloran Shift Control Engineer		
OKONKWO, Dessa Maintenance Technician		
WRENN, Piotr RDSI officer		
MBEKI, Cato Assistant Operator		

MOTIVE

AUTHORITIES

ENTITLED TO	SPENT
NUMBERS USED	UNSPENT

ACCUSATIONS

NO.	NAMED	TIME	OUTCOME
1			
2			
3 —			
FINAL			

FINDING OF THE REVIEW

The deaths of Aurelio Sarn and Ines Ballard are found to be:
☐ MURDER-SUICIDE, AS ORIGINALLY FOUND ☐ DOUBLE UNLAWFUL KILLING ☐ OTHER (state)

If unlawful killing by a third party, that person is:

SIGNED, LEAD REVIEWER TIME

SOLO PLAY — PHASED RELEASE

No Conference. Five Authorities. Open one file per phase, in this order, and write a hypothesis before opening the next.

PHASE	OPEN	BEFORE MOVING ON, WRITE DOWN
1	Case Board only	Which of the two deaths came first, and why you think so
2	+ File F-2 (Hydrology)	A time window for Ballard entering the water
3	+ File F-1 (Control)	Who could have been at console 1 at 06:02
4	+ File F-4 (Security)	Where her credential card was, and who could reach it
5	+ Files F-3 and F-5	— accuse

Authorities may be spent from Phase 2 onward.

NOW OPEN ENVELOPE S-1